Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McCaffery, Peter John Ph.D.		
Project Title:	The Effect of Altered Gravity on Retinoic Acid Signaling in the Developing Embryo		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Waltham	**State:**	MA
Zip Code:	02452	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1438		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Pregnant animals between embryonic days 3-11 will be subjected to 2 g hypergravity in the 24-foot centrifuge at the NASA-Ames Research Center and analyzed for possible alterations in Retinoic Acid (RA) signaling. The analysis of gene expression, by in-situ hybridization, will start at embryonic day 9, giving 6 days for the animals to adapt to centrifugation. At this time point we will assay for heat shock genes activated by stress to determine the extent to which the animals have recovered from the initial stress. The tests for retinoic acid regulated genes will be at embryonic days 11 and 13. This will be on genes that are not normally transcribed until these later ages and will be unaffected by the initial stress over the first 4-5 days. Specifically we will look at changes in transcript expression of COUPTFII, sonic hedgehog, Eph B2, ephrin B1, and ephrin B2.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is an investigation into the effects of hypergravity on the early development of the mouse embryonic central nervous system (CNS), with an emphasis on transcriptional regulation by retinoic acid. Retinoic acid controls the expression of a large number of growth factors and receptors which regulate CNS development. Several embryonic developmental defects, caused by space flight or gravitational changes, bear a significant similarity to the defects resulting from retinoid signaling abnormalities. These studies delve into the mechanisms by which developmental abnormalities may occur when the embryo is exposed to hypergravity, with the assumption that the effects of hypergravity may, at least in part, be due to changes in retinoic acid controlled transcription. This work will provide insight into the basic mechanism by which altered gravity may influence gene transcription as well as provide a new slant to the view of retinoic acid as a teratogen, possibly revealing new aspects of the selectivity of the compound to specific regions of the developing brain.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)